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Report:

Design of a Heavy Duty Parallel Shafts Double Reduction Gearbox

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Abstract

This report presents the design and analysis of a heavy-duty, parallel-shaft double reduction gearbox intended for a rated input power of 175 kW at 1450 rpm with a total reduction ratio of 1:10. While helical gears are the industry standard for high-power transmission, this design utilizes a spur gear configuration to align with available manufacturing capabilities, specifically large-scale casting. The selection of spur gears eliminates the generation of axial thrust forces, thereby removing the requirement for thrust bearings. Thermal management is addressed through an integrated active cooling system, comprising a heat exchanger and pump, ensuring optimal oil operating temperatures despite the high power throughput.

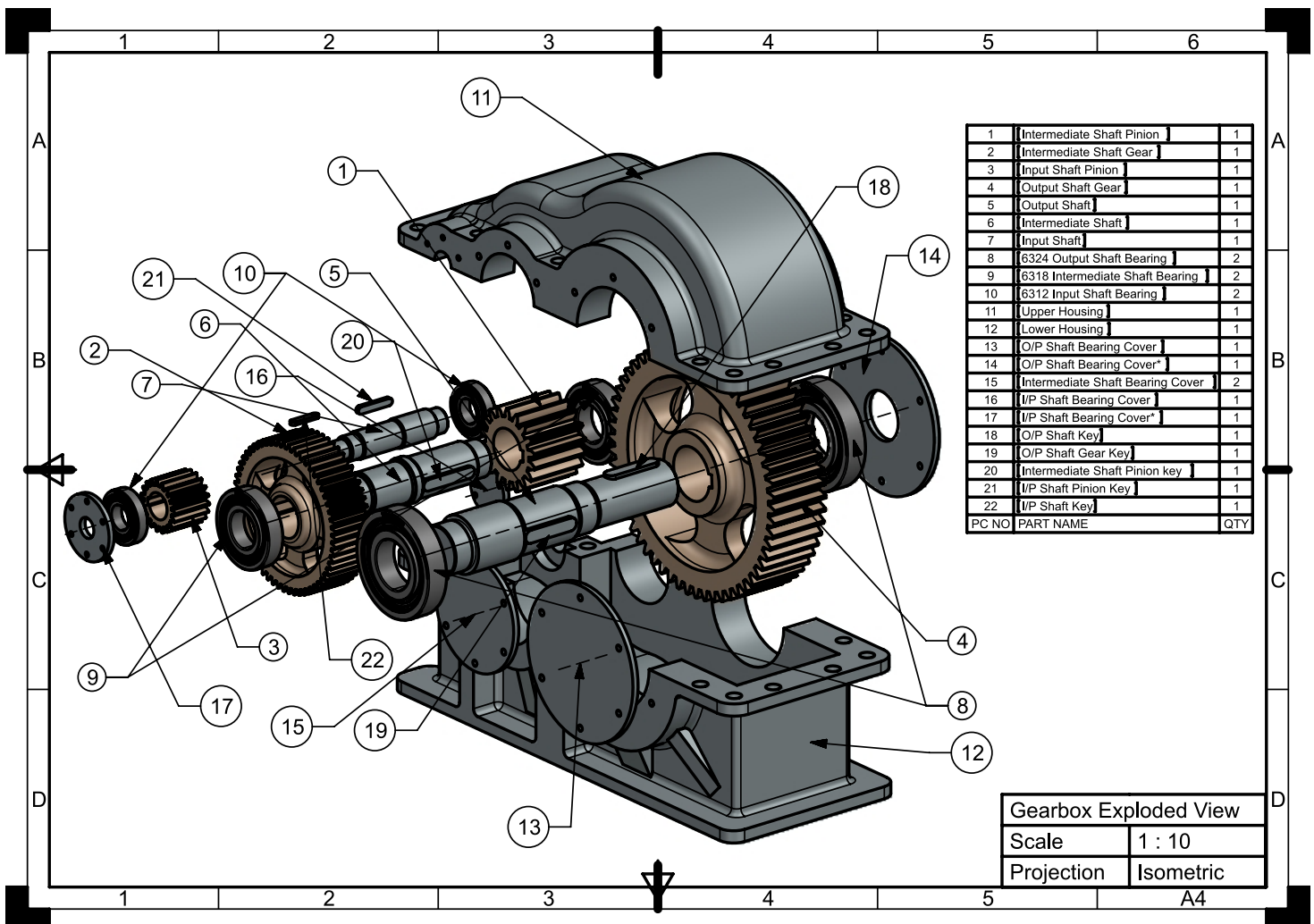


Figure 1: Gearbox Assembly List of Parts

1. Introduction

The objective of this project is to design a robust speed reduction unit capable of transmitting high torque for heavy-duty applications. The system specifications require a reduction ratio of 1:10 with an input power of 175 kW.

A critical aspect of this design process was the selection of the gear type. Although helical gears are typically preferred for their smooth operation and load-carrying capacity, spur gears were selected for this specific application. This decision was driven by three key factors:

- **Manufacturing Constraints:** The available manufacturing facilities rely on casting processes best suited for spur gear profiles.
- **Operational Environment:** The application is situated in an environment without strict noise regulations, mitigating the primary disadvantage of spur gears.
- **Assembly Simplicity:** The use of spur gears eliminates axial thrust loads, simplifying the bearing arrangement and housing design.

To ensure reliability, thermal concerns are mitigated by an external forced lubrication and cooling system. This report details the workflow being used, design calculations, component selection, CAD modeling, and structural Finite Element Analysis simulation.

Category	Specification	Design Constraint
Input Conditions	175 kW @ 1450 rpm	Heavy-Duty Application
Kinematics	Target Ratio 1:10	Double Reduction Stage
Gear Profile	Spur Gears	Suited for Casting Facilities
Thermal Control	Forced Lubrication	External Cooling System
Environment	Noise Tolerant Zone	No Axial Thrust Loads

Table 1: Project Design Specifications and Constraints

2. Design Workflow

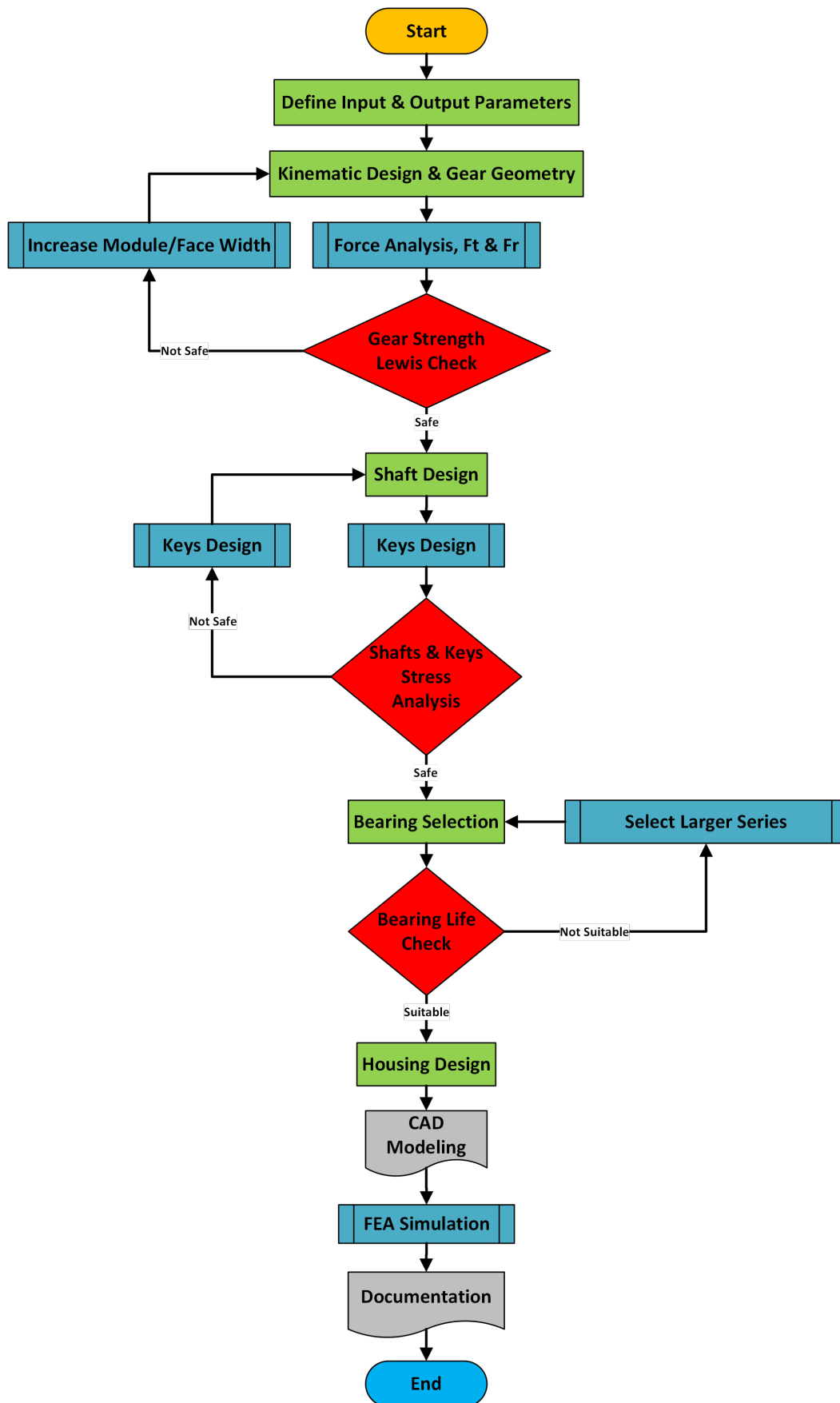


Figure 2: Design Workflow Flowchart Diagram

2.1 Kinematic Analysis

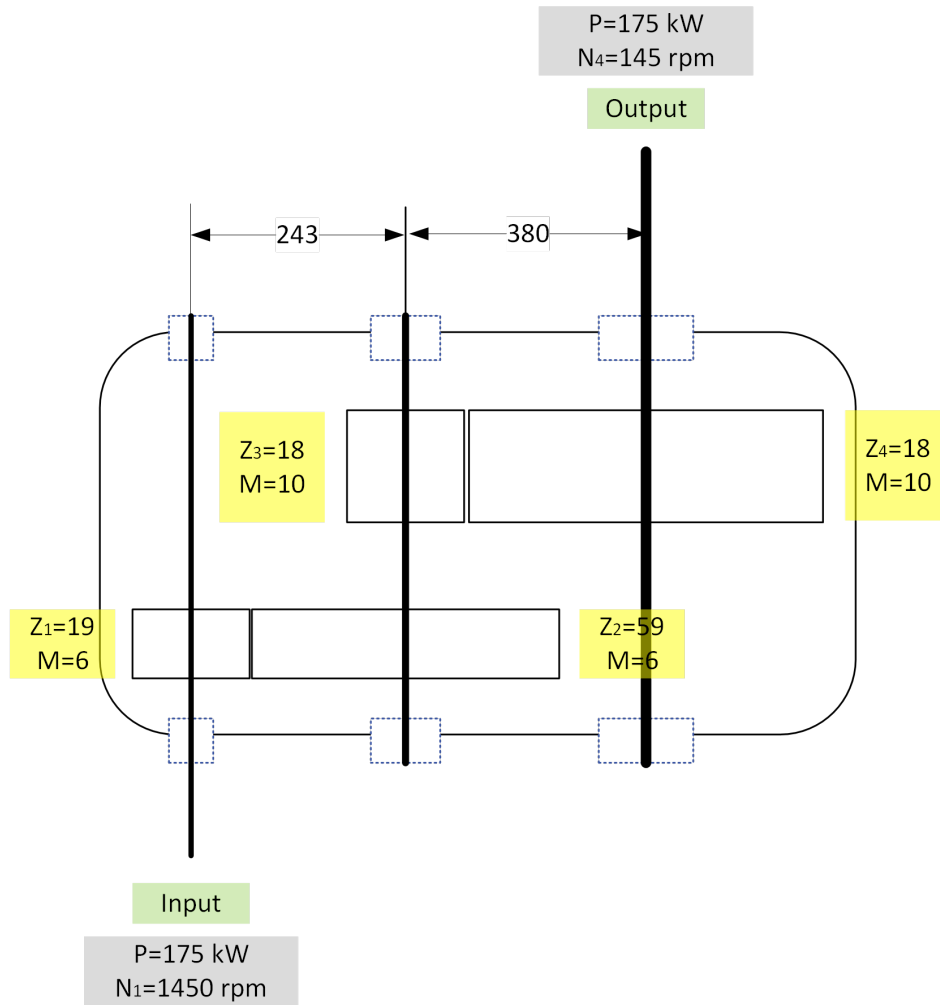


Figure 3: Kinematic Diagram of the Gearbox

The total reduction ratio (i_{total}) is calculated as the product of the individual stage ratios:

$$i_1 = \frac{z_2}{z_1} = \frac{59}{19} \approx 3.105 \quad (1)$$

$$i_2 = \frac{z_4}{z_3} = \frac{18}{18} \approx 1.000 \quad (2)$$

$$i_{total} = i_1 \times i_2 = 3.105 \times 1.000 = 3.105 \quad (3)$$

2.2 Speed and Torque Analysis

Given an input power $P = 175$ kW and input speed $n_{in} = 1450$ rpm, the torque at each shaft is calculated using:

$$T = \frac{2\pi \times P \times 10^3}{n}$$

1. Input Shaft (Shaft I):

$$T_1 = \frac{2\pi \times 10^3 \times 175}{1450} = 1,152.6 \text{ N} \cdot \text{m} \quad (4)$$

2. Intermediate Shaft (Shaft II):

$$n_{inter} = \frac{n_{in}}{i_1} = \frac{1450}{3.105} = 467 \text{ rpm} \quad (5)$$

$$T_2 = \frac{2\pi \times 10^3 \times 175}{467} = 3,579.2 \text{ N} \cdot \text{m} \quad (6)$$

3. Output Shaft (Shaft III):

$$n_{out} = \frac{n_{inter}}{i_2} = \frac{467}{3.222} = 145 \text{ rpm} \quad (7)$$

$$T_3 = \frac{2\pi \times 10^3 \times 175}{145} = 11,526 \text{ N} \cdot \text{m} \quad (8)$$

2.3 Gear Force Analysis

The tangential force (F_t), the radial force (F_r) and the total resultant load (R) acting on the gear teeth are calculated assuming a pressure angle of $\alpha = 20^\circ$.

Stage 1 (Pinion 1, $m = 6$, $z_1 = 19$):

$$d_{p1} = m \cdot z_1 = 6 \cdot 19 = 114 \text{ mm} = 0.114 \text{ m} \quad (9)$$

$$F_{t1} = \frac{2 \cdot T_1}{d_{p1}} = \frac{2 \cdot 1152.6}{0.114} = 20,221 \text{ N} \approx 20.2 \text{ kN} \quad (10)$$

$$F_{r1} = F_{t1} \cdot \tan(\alpha) = 20,221 \cdot \tan(20^\circ) = 7,360 \text{ N} \approx 7.4 \text{ kN} \quad (11)$$

$$R_1 = \frac{F_{t1}}{\cos(\alpha)} = \frac{20,221}{\cos(20^\circ)} = 21,519 \text{ N} \approx 21.5 \text{ kN} \quad (12)$$

Stage 2 (Pinion 2, $m = 10$, $z_3 = 18$):

$$d_{p3} = m \cdot z_3 = 10 \cdot 18 = 180 \text{ mm} = 0.18 \text{ m} \quad (13)$$

$$F_{t2} = \frac{2 \cdot T_2}{d_{p3}} = \frac{2 \cdot 3579.2}{0.18} = 39,769 \text{ N} \approx 39.8 \text{ kN} \quad (14)$$

$$F_{r2} = F_{t2} \cdot \tan(\alpha) = 39,769 \cdot \tan(20^\circ) = 14,475 \text{ N} \approx 14.5 \text{ kN} \quad (15)$$

$$R_2 = \frac{F_{t2}}{\cos(\alpha)} = \frac{39,769}{\cos(20^\circ)} = 42,321 \text{ N} \approx 42.3 \text{ kN} \quad (16)$$

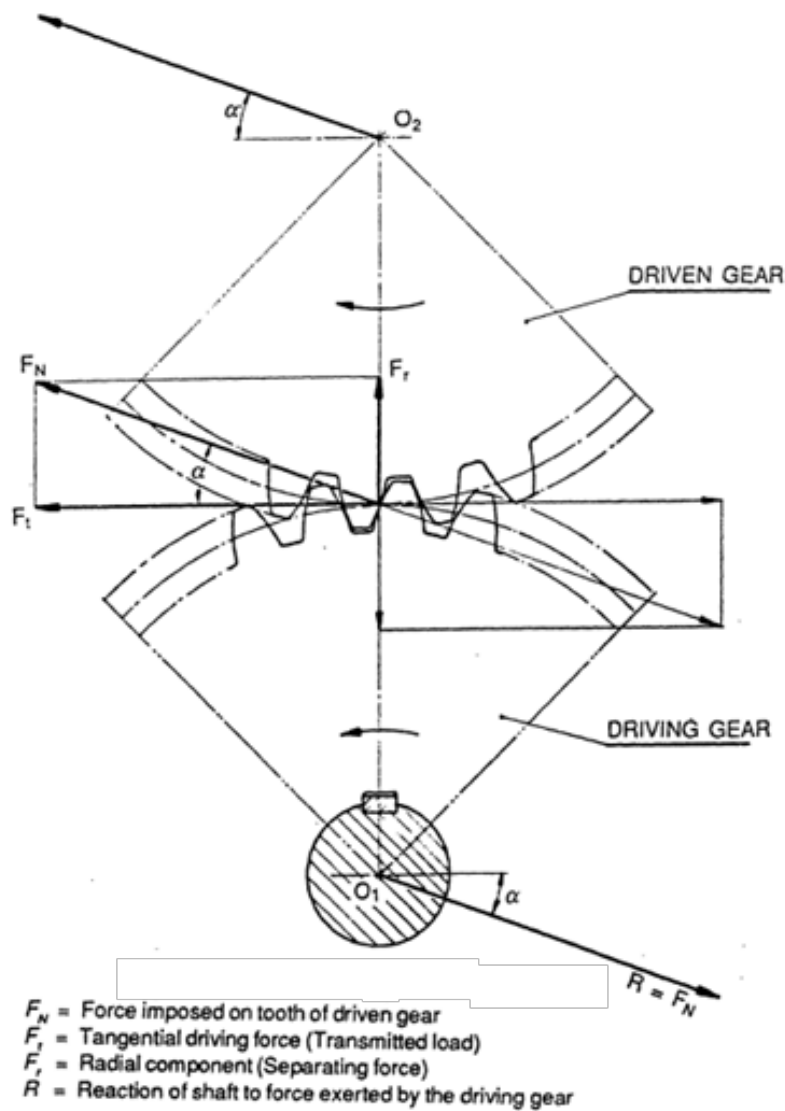


Figure 4: Force analysis on a spur gear

The tangential load doubles from Stage 1 to Stage 2 (20.2 kN $\rightarrow$ 39.8 kN). Therefore, increasing the module to $m = 10$ is necessary to maintain tooth bending strength.

2.4 Lewis Bending Stress Check

The bending stress (σ_b) acting on the gear teeth is calculated using the Lewis formula:

$$\sigma_b = \frac{F_t}{b \cdot m \cdot Y} \quad (17)$$

Where: F_t : Tangential force (N), b : Face width (mm), m : Module (mm), Y : Lewis form factor

Stage 1 (Pinion 1):

- $F_{t1} = 20,221 \text{ N}$
- $b_1 = 100 \text{ mm}$
- $m = 6 \text{ mm}$
- $z_1 = 19$

First, the Lewis Form Factor (Y_1):

$$Y_1 = 0.314 @ z_1 = 19 \quad (18)$$

Now, calculate the induced bending stress (σ_{b1}):

$$\sigma_{b1} = \frac{20,221}{100 \cdot 6 \cdot 0.314} = \frac{20,221}{199.8} \approx 101.21 \text{ MPa} \quad (19)$$

Stage 2 (Pinion 2):

- $F_{t2} = 39,769 \text{ N}$
- $b_2 = 150 \text{ mm}$
- $m = 10 \text{ mm}$
- $z_3 = 18$

The Lewis Form Factor (Y_2):

$$Y_1 = 0.309 @ z_2 = 18 \quad (20)$$

Calculate the induced bending stress (σ_{b2}):

$$\sigma_{b2} = \frac{39,769}{150 \cdot 10 \cdot 0.309} = \frac{39,769}{487.5} \approx 81.58 \text{ MPa} \quad (21)$$

2.5 Material Selection

Given the manufacturing requirement for casting and the induced bending stresses calculated previously, **Cast Steel (ASTM A148 Grade 80-50)** is selected. Unlike grey cast iron, cast steel provides the necessary ductility and yield strength to withstand the calculated loads without brittle failure.

Material Properties (ASTM A148 Grade 80-50):

- **Ultimate Tensile Strength (S_{ut}):** 550 MPa
- **Yield Strength (S_y):** 345 MPa
- **Allowable Bending Stress (σ_{all}):** estimated as $0.7 \cdot S_y$:

$$\sigma_{all} \approx 0.7 \cdot 345 \text{ MPa} = 245 \text{ MPa} \quad (22)$$

2.6 Safety Factor Verification

The Factor of Safety (n) is calculated by comparing the allowable stress of the selected cast steel to the induced Lewis bending stress (σ_b) for each stage.

Stage 1 Safety Check:

$$n_1 = \frac{\sigma_{all}}{\sigma_{b1}} = \frac{245 \text{ MPa}}{101.21 \text{ MPa}} \approx 2.42 \quad (23)$$

Since $n_1 > 1.5$, the design is **Safe** for Stage 1.

Stage 2 Safety Check:

$$n_2 = \frac{\sigma_{all}}{\sigma_{b2}} = \frac{245 \text{ MPa}}{81.58 \text{ MPa}} \approx 3.0 \quad (24)$$

Since $n_2 > 1.5$, the design is **Safe** for Stage 2.

The Lewis formula is used here as a simplified static baseline, where its omission of dynamic and stress concentration factors is compensated for by obtaining a higher

Factor of Safety of 2.5, ensuring robustness comparable to the more complex AGMA method.

Stage	Module	Teeth (z_p/z_g)	Face Width	Bending Stress	Safety Factor
Stage 1	6 mm	19 / 59	100 mm	101.21 MPa	2.42
Stage 2	10 mm	18 / 58	150 mm	81.58 MPa	3.00

Table 2: Gear Geometry and Safety Factor Summary (Material: ASTM A148 80-50)

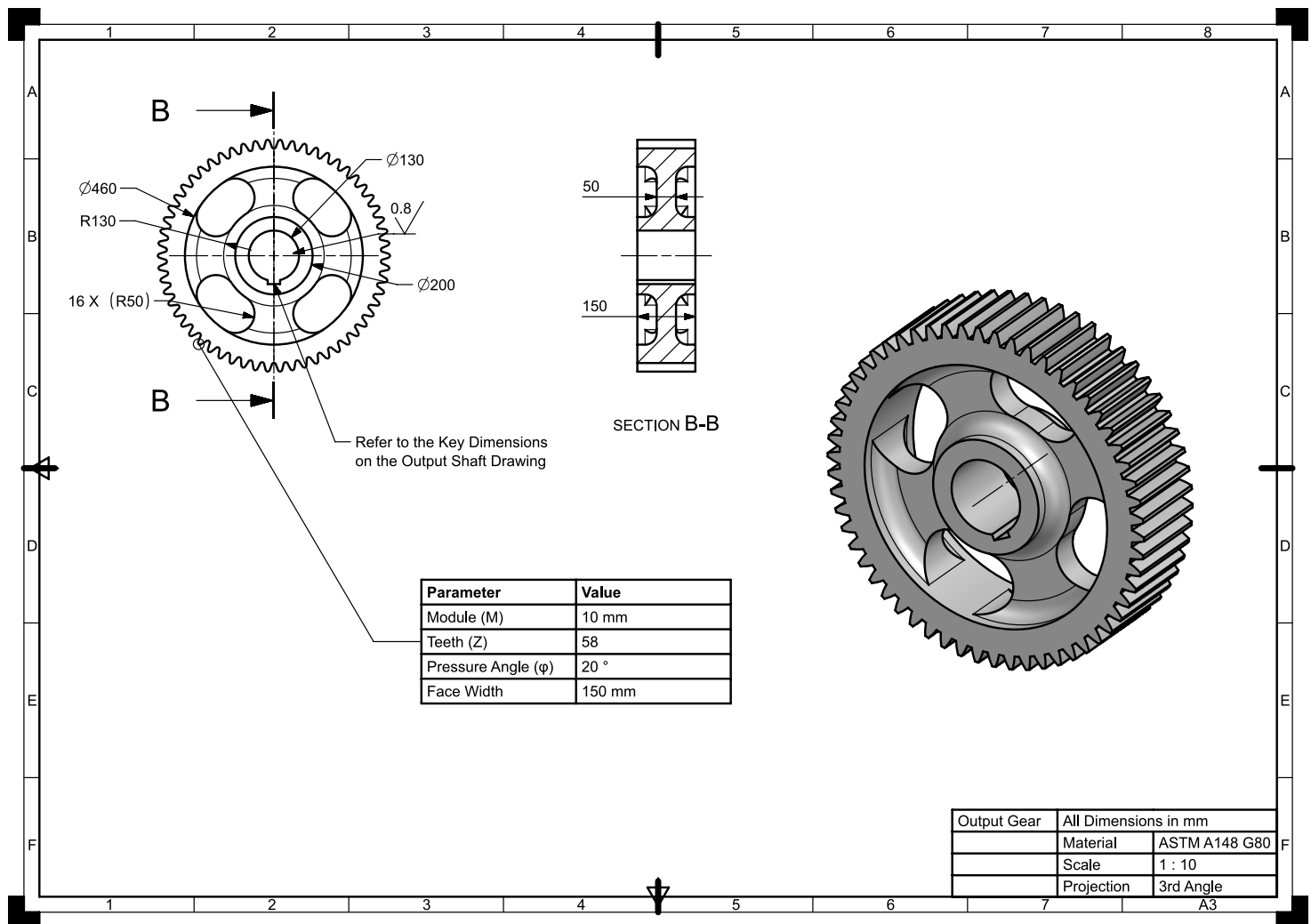


Figure 5: Drawing of the Output Gear

2.7 Shaft Torsional Stress Verification

The shear stress due to torsion is calculated as

$$\tau = \frac{16T}{\pi d^3}$$

Input Shaft ($d = 50 \text{ mm}$):

$$\tau_1 = \frac{16 \cdot 1152.6}{\pi \cdot (0.050)^3} = \frac{18,441.6}{0.0003927} = 46.9 \text{ MPa} \quad (25)$$

Intermediate Shaft ($d = 90 \text{ mm}$):

$$\tau_2 = \frac{16 \cdot 3579.2}{\pi \cdot (0.090)^3} = \frac{57,267.2}{0.00229} = 25.0 \text{ MPa} \quad (26)$$

Output Shaft ($d = 110 \text{ mm}$):

$$\tau_3 = \frac{16 \cdot 11,526}{\pi \cdot (0.110)^3} = \frac{184,416}{0.00418} = 44.1 \text{ MPa} \quad (27)$$

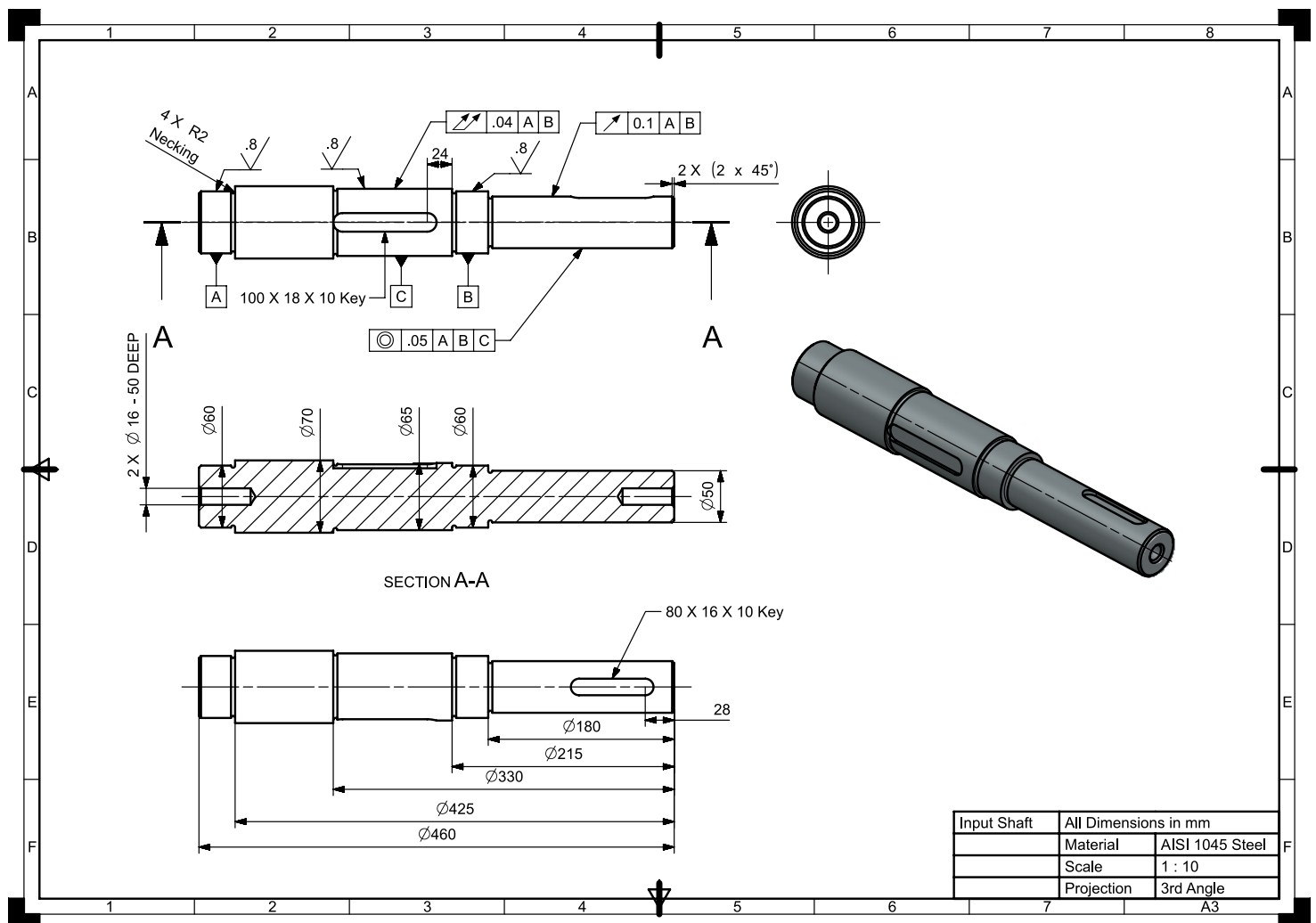


Figure 6: GD&T Drawing of the Input Shaft ready to release & Manufacturing

2.8 Material Selection and Factor of Safety

The material selected for the shafts is **AISI 1045 Cold Drawn Steel**.

- Yield Strength (S_y): 530 MPa
- Shear Yield Strength (S_{sy}): According to the Distortion Energy (Von Mises) theory,
 $S_{sy} = 0.577S_y$.

$$S_{sy} = 0.577 \cdot 530 = 305.8 \text{ MPa} \quad (28)$$

The minimum factor of safety (n) is calculated based on the maximum torsional stress occurring in the Input Shaft ($\tau_{max} = 46.9 \text{ MPa}$):

$$n = \frac{S_{sy}}{\tau_{max}} = \frac{305.8}{46.9} = 6.52 \quad (29)$$

Shaft	Diameter	Material	Shear Strength (S_{sy})	Torsional Stress
Input (I)	50 mm	AISI 1045	305.8 MPa	46.9 MPa
Intermediate (II)	90 mm	AISI 1045	305.8 MPa	25.0 MPa
Output (III)	110 mm	AISI 1045	305.8 MPa	44.1 MPa

Table 3: Shaft Dimensions and Torsional Stress Summary

2.9 Von Mises Stress Analysis (Combined Loading)

To ensure the shafts can withstand the combined effects of torsional shear stress and bending moments (due to gear forces), the equivalent Von Mises stress (σ') was analyzed using SOLIDWORKS Simulation Premium. The failure criterion is defined such that yielding occurs when the equivalent stress exceeds the yield strength of the material.

The stress distribution results for the shafts are presented below:

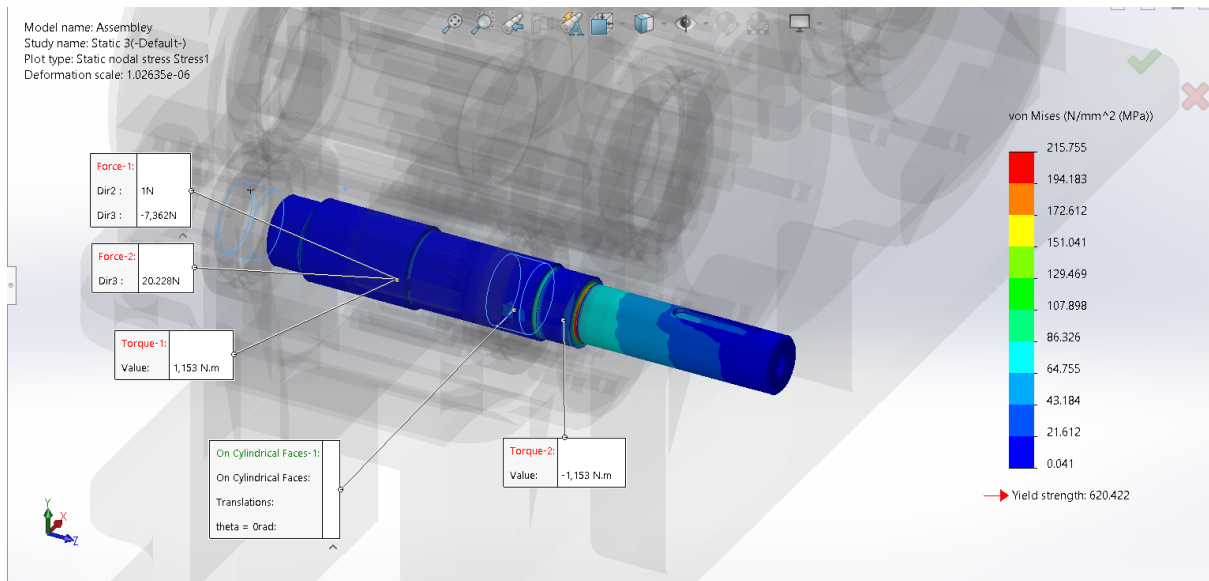


Figure 7: Von Mises Stress Distribution - Input Shaft

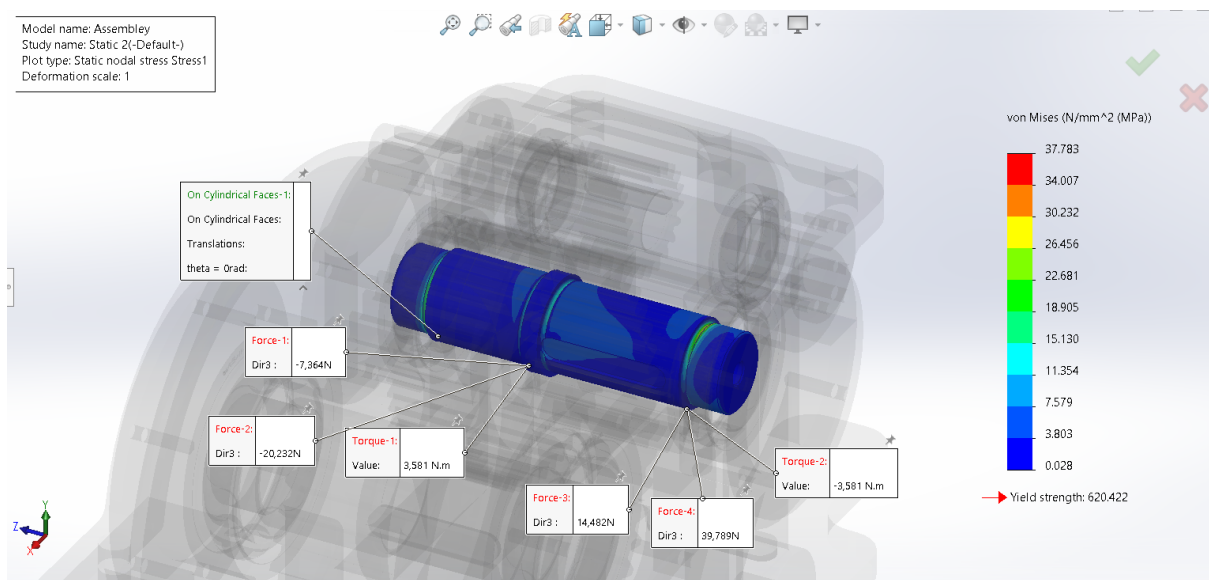


Figure 8: Von Mises Stress Distribution - Intermediate Shaft

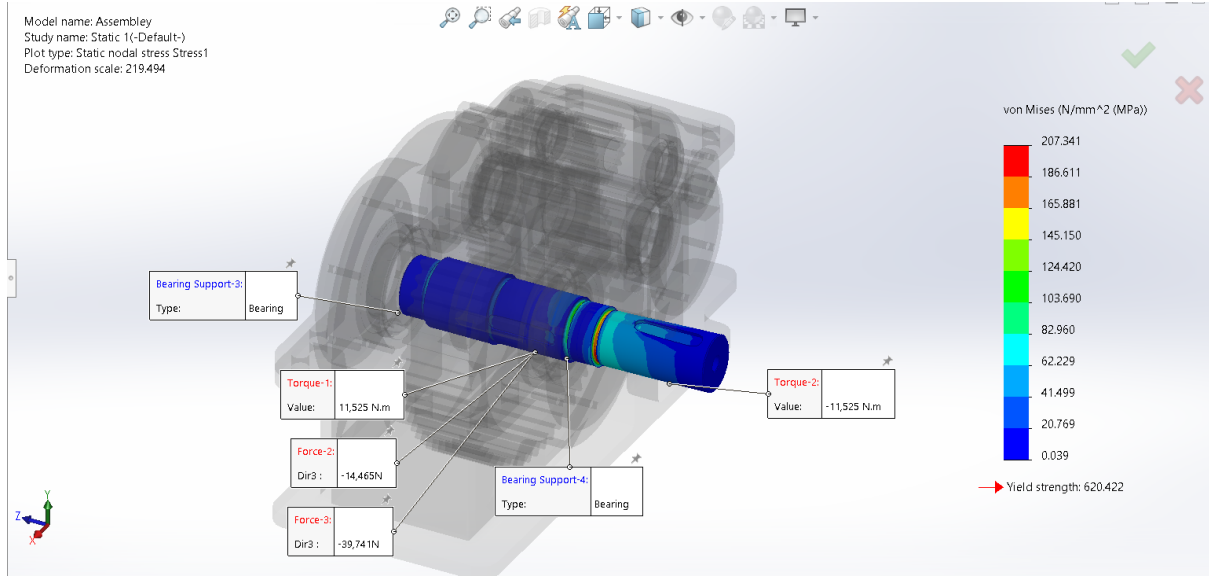


Figure 9: Von Mises Stress Distribution - Output Shaft

2.9.1 Final Safety Verification

Based on the analysis, the maximum Von Mises stress (σ'_{max}) observed in the system is **215 MPa**.

Using the previously selected material, **AISI 1045 Cold Drawn Steel** ($S_y = 530$ MPa), the final Factor of Safety (n) is calculated as:

$$n = \frac{S_y}{\sigma'_{max}} = \frac{530}{215} = \mathbf{2.46} \quad (30)$$

Since $n > 1$, the shaft design is verified to be safe under combined loading conditions.

2.10 Key Stress Verification

The keys are verified for two failure modes: Shear failure (τ) and Crushing failure (σ_c).

$$\tau = \frac{2T}{d \cdot b \cdot L} \quad \text{and} \quad \sigma_c = \frac{4T}{d \cdot h \cdot L}$$

2.10.1 1. Input Shaft (Torque = 1152.6 Nm)

A. Input Coupling Key ($80 \times 16 \times 10$ mm on $\phi 50$ shaft):

$$\tau_{k1} = \frac{2 \cdot 1152.6}{0.050 \cdot 0.016 \cdot 0.080} = 36.0 \text{ MPa} \quad (31)$$

$$\sigma_{c1} = \frac{4 \cdot 1152.6}{0.050 \cdot 0.010 \cdot 0.080} = 115.3 \text{ MPa} \quad (32)$$

B. Input Gear Key (100 × 18 × 10 mm on $\phi 65$ shaft):

$$\tau_{k2} = \frac{2 \cdot 1152.6}{0.065 \cdot 0.018 \cdot 0.100} = 19.7 \text{ MPa} \quad (33)$$

$$\sigma_{c2} = \frac{4 \cdot 1152.6}{0.065 \cdot 0.010 \cdot 0.100} = 70.9 \text{ MPa} \quad (34)$$

2.10.2 2. Intermediate Shaft (Torque = 3579 Nm)

A. Stage 1 Gear Key (100 × 28 × 14 mm on $\phi 100$ shaft):

$$\tau_{k3} = \frac{2 \cdot 3579}{0.100 \cdot 0.028 \cdot 0.100} = 25.6 \text{ MPa} \quad (35)$$

$$\sigma_{c3} = \frac{4 \cdot 3579}{0.100 \cdot 0.014 \cdot 0.100} = 102.3 \text{ MPa} \quad (36)$$

B. Stage 2 Pinion Key (150 × 28 × 14 mm on $\phi 100$ shaft):

$$\tau_{k4} = \frac{2 \cdot 3579}{0.100 \cdot 0.028 \cdot 0.150} = 17.0 \text{ MPa} \quad (37)$$

$$\sigma_{c4} = \frac{4 \cdot 3579}{0.100 \cdot 0.014 \cdot 0.150} = 68.2 \text{ MPa} \quad (38)$$

2.10.3 3. Output Shaft (Torque = 11,526 Nm)

A. Output Gear Key (150 × 32 × 16 mm on $\phi 130$ shaft):

$$\tau_{k5} = \frac{2 \cdot 11526}{0.130 \cdot 0.032 \cdot 0.150} = 36.9 \text{ MPa} \quad (39)$$

$$\sigma_{c5} = \frac{4 \cdot 11526}{0.130 \cdot 0.016 \cdot 0.150} = 147.8 \text{ MPa} \quad (40)$$

B. Output Coupling Key (150 × 32 × 16 mm on $\phi 110$ shaft):

$$\tau_{k6} = \frac{2 \cdot 11526}{0.110 \cdot 0.032 \cdot 0.150} = 43.7 \text{ MPa} \quad (41)$$

$$\sigma_{c6} = \frac{4 \cdot 11526}{0.110 \cdot 0.016 \cdot 0.150} = 174.6 \text{ MPa} \quad (42)$$

2.11 Material Selection and Safety Verification

The keys are selected to be manufactured from **C45 Cold Drawn Steel**.

- **Yield Strength (S_y):** 370 MPa
- **Allowable Shear Stress (τ_{all}):** $0.5S_y = 185$ MPa
- **Allowable Crushing Stress ($\sigma_{c,all}$):** $\approx S_y = 370$ MPa

Safety Factor (S.F.) Calculation (Based on Critical Output Key):

$$S.F._{shear} = \frac{\tau_{all}}{\tau_{max}} = \frac{185}{43.7} = 4.23 \quad (> 2.0 \text{ Safe}) \quad (43)$$

$$S.F._{crushing} = \frac{\sigma_{c,all}}{\sigma_{c,max}} = \frac{370}{174.6} = 2.12 \quad (> 2.0 \text{ Safe}) \quad (44)$$

All keys are safe against both shear and crushing failure.

2.12 Bearing Selection and Life Analysis

The bearings were selected to support the radial and axial loads generated by the gear forces. The reaction forces at the bearings are calculated based on the static equilibrium of the shafts, treating them as simply supported beams. The bearing life is estimated using the standard L_{10h} formula for ball bearings:

$$L_{10h} = \frac{10^6}{60 \cdot n} \left(\frac{C}{K_s \cdot P} \right)^3 \quad (45)$$

Where:

- L_{10h} : Basic rating life (hours)
- n : Rotational speed (rpm)
- C : Basic dynamic load rating (kN)
- P : Equivalent dynamic bearing load (kN)
- K_s : Service factor

2.12.1 1. Input Shaft Bearing Analysis (Shaft I)

Geometry & Loading:

- **Bearing Span (L):** 241 mm
- **Load Position:** The pinion load acts at 80.5 mm from Bearing (B).
- **Distances:** $a = 160.5$ mm (from Bearing A), $b = 80.5$ mm (from Bearing B).
- **Forces:** $F_{t1} = 20,221$ N, $F_{r1} = 7,360$ N.

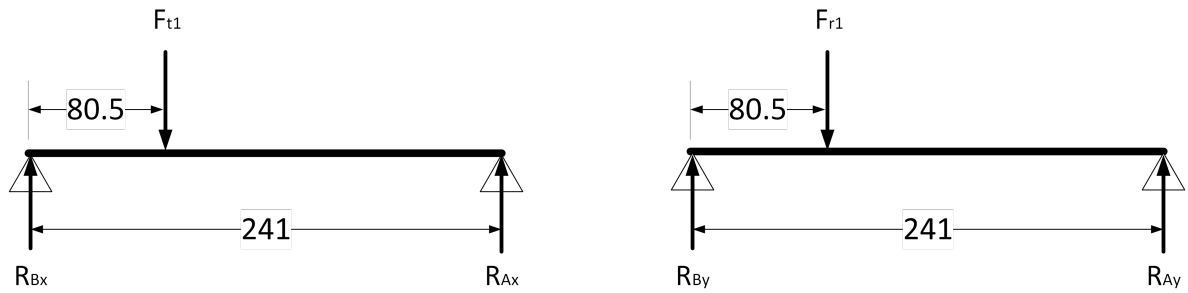


Figure 10: Reaction Forces on the input Shaft

Reaction Calculations: Since the load is closer to Bearing (B), it supports the majority of the load.

$$R_{By} = \frac{F_{r1} \cdot a}{L} = \frac{7,360 \cdot 160.5}{241} = 4,901 \text{ N} \quad (46)$$

$$R_{Bx} = \frac{F_{t1} \cdot a}{L} = \frac{20,221 \cdot 160.5}{241} = 13,467 \text{ N} \quad (47)$$

$$R_B = \sqrt{R_{Bx}^2 + R_{By}^2} = \sqrt{13,467^2 + 4,901^2} = 14,332 \text{ N} \approx 14.33 \text{ kN} \quad (48)$$

Bearing (A) supports the remainder:

$$R_A = \sqrt{(20,221 - 13,467)^2 + (7,360 - 4,901)^2} = \sqrt{6,754^2 + 2,459^2} \approx 7.19 \text{ kN} \quad (49)$$

Life Calculation (Selected: 6312):

- **Designation:** 6312 Deep Groove Ball Bearing
- **Dynamic Load Rating (C):** 135 kN
- **Maximum Equivalent Load (P):** 14 kN (At Bearing B)

- **Speed (n):** 1450 rpm

$$L_{10h} = \frac{10^6}{60 \cdot 1450} \left(\frac{135}{14} \right)^3 = 11.49 \cdot (9.64)^3 \approx \mathbf{10,320 \text{ hours}} \quad (50)$$

2.12.2 2. Intermediate Shaft Bearing Analysis (Shaft II)

Geometry & Loading:

- **Bearing Span (L):** 341.5 mm
- **Gear 2 (Input):** Located 86.5 mm from Bearing (B). ($F_{t1} = 20.2 \text{ kN}$, $F_{r1} = 7.4 \text{ kN}$).
- **Pinion 3 (Output):** Located 110 mm from Bearing (A). ($F_{t2} = 39.8 \text{ kN}$, $F_{r2} = 14.5 \text{ kN}$).

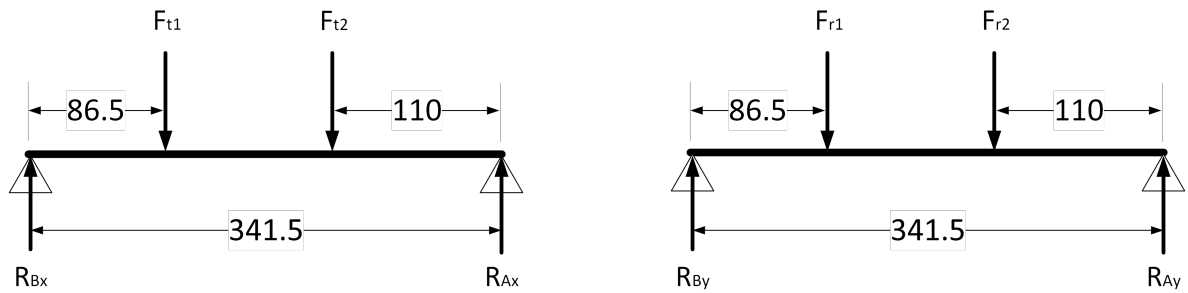


Figure 11: Reaction Forces on the intermediate Shaft

Reaction Calculations (Tangential Plane): Summing moments about Bearing A, where Gear 2 is at distance $341.5 - 86.5 = 255 \text{ mm}$:

$$\sum M_A = 0 \Rightarrow R_{Bx(tang)} \cdot 341.5 + F_{t2} \cdot 110 - F_{t1} \cdot 255 = 0 \quad (51)$$

$$R_{Bx(tang)} \cdot 341.5 = (20,221 \cdot 255) - (39,769 \cdot 110) = 5,156,355 - 4,374,590 = 781,765 \quad (52)$$

$$R_{Bx(tang)} = \frac{781,765}{341.5} = 2,290 \text{ N} \quad (53)$$

Summing forces:

$$R_{Ax(tang)} = |F_{t1} - F_{t2} - R_{Bx(tang)}| = |20,221 - 39,769 - 2,290| = 21,838 \text{ N} \quad (54)$$

Reaction Calculations (Radial Plane): Radial forces (F_r) always act towards the shaft center (separating forces), so they add up:

$$\sum M_A = 0 \Rightarrow R_{By(rad)} \cdot 341.5 - F_{r2} \cdot 110 - F_{r1} \cdot 255 = 0 \quad (55)$$

$$R_{By(rad)} = \frac{(14,475 \cdot 110) + (7,360 \cdot 255)}{341.5} = \frac{3,469,050}{341.5} = 10,158 \text{ N} \quad (56)$$

$$R_{Ay(rad)} = (F_{r1} + F_{r2}) - R_{By(rad)} = (7,360 + 14,475) - 10,158 = 11,677 \text{ N} \quad (57)$$

Resultant Loads:

$$R_A = \sqrt{21,838^2 + 11,677^2} = 24,765 \text{ N} \approx \mathbf{24.8 \text{ kN}} \quad (\text{Max Load}) \quad (58)$$

$$R_B = \sqrt{2,290^2 + 10,158^2} = 10,412 \text{ N} \approx 10.4 \text{ kN} \quad (59)$$

Life Calculation (Selected: 6318):

- **Designation:** 6318 Deep Groove Ball Bearing
- **Dynamic Load Rating (C):** 178 kN
- **Speed (n):** 467 rpm

$$L_{10h} = \frac{10^6}{60 \cdot 467} \left(\frac{178}{24.8} \right)^3 = 35.69 \cdot (7.18)^3 \approx \mathbf{12,950 \text{ hours}} \quad (60)$$

2.12.3 3. Output Shaft Bearing Analysis (Shaft III)

Geometry & Loading:

- **Bearing Span (L):** 353.5 mm
- **Load Position:** Gear 4 is located 117.5 mm from Bearing (A).
- **Forces:** $F_{t2} = 39,769 \text{ N}$, $F_{r2} = 14,475 \text{ N}$.

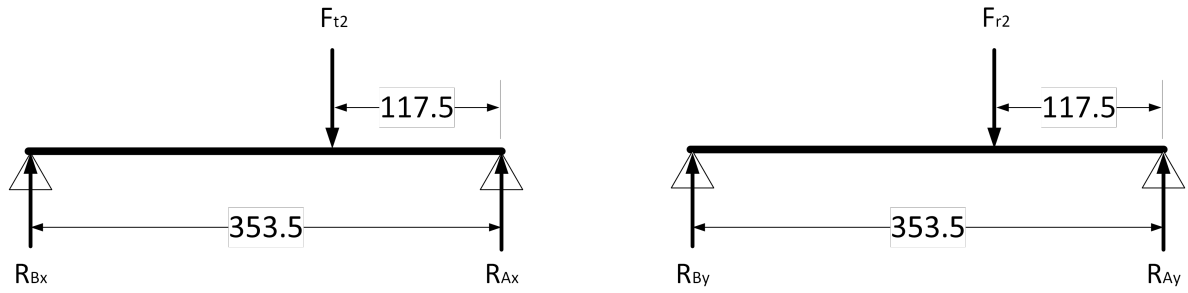


Figure 12: Reaction Forces on the output Shaft

Reaction Calculations: Bearing (A) is closer to the load, sustains maximum reaction.

$$R_{Ax} = F_{t2} \cdot \frac{(353.5 - 117.5)}{353.5} = 39,769 \cdot \frac{236}{353.5} = 26,550 \text{ N} \quad (61)$$

$$R_{Ay} = F_{r2} \cdot \frac{236}{353.5} = 14,475 \cdot 0.6676 = 9,663 \text{ N} \quad (62)$$

$$R_A = \sqrt{26,550^2 + 9,663^2} = 28,252 \text{ N} \approx \mathbf{28.3 \text{ kN}} \quad (63)$$

Life Calculation (Selected: 6324):

- **Designation:** 6324 Deep Groove Ball Bearing
- **Dynamic Load Rating (C):** 208 kN
- **Speed (n):** 145 rpm

$$L_{10h} = \frac{10^6}{60 \cdot 145} \left(\frac{208}{28.3} \right)^3 = 114.9 \cdot (7.35)^3 \approx \mathbf{45,600 \text{ hours}} \quad (64)$$

2.13 Bearing Selection Summary

Shaft	Bearing	Load Rating (C)	Max Load (P)	Life (L_{10h})
Input	6312	135 kN	14.3 kN	10,320 hrs
Intermediate	6318	178 kN	24.8 kN	12,950 hrs
Output	6324	208 kN	28.3 kN	45,600 hrs

Table 4: Summary of Bearing Selections and Calculated Life

2.14 Housing Design and Structural Verification

The gearbox housing was designed to provide rigid support for the shafts, maintain precise gear alignment, and contain the lubricant. The structural integrity of the housing was verified through Finite Element Analysis (FEA).

2.14.1 1. Material Selection

Grey Cast Iron (ASTM A48 Class 30) was selected for the housing due to its excellent vibration damping properties, castability, and high compressive strength.

- **Yield Strength (S_y):** ≈ 200 MPa (Tension), 750 MPa (Compression)
- **Young's Modulus (E):** 110 GPa
- **Poisson's Ratio (ν):** 0.26

2.14.2 2. Initial Dimensioning (Empirical Calculation)

The minimum wall thickness (t) of the housing was determined using standard empirical relations for industrial gearboxes to ensuring sufficient stiffness before optimization.

Using the relation based on the maximum output torque ($T_{max} = 11,526 \text{ N} \cdot \text{m}$):

$$t_{min} = 0.03 \cdot \sqrt[3]{T_{max} \cdot 1000} + 5 \text{ mm} \quad (65)$$

$$t_{min} = 0.03 \cdot \sqrt[3]{11,526,000} + 5 = 0.03 \cdot 225.8 + 5 \approx 11.77 \text{ mm} \quad (66)$$

To ensure robustness and account for casting tolerances, a design wall thickness of **20 mm** was selected, with reinforced bearing seats of **60 mm**.

2.14.3 3. FEA Simulation Setup

A static structural analysis was performed to verify the housing under maximum loading conditions.

- **Boundary Conditions:** The base of the housing was fixed.

- **Loads:** The calculated bearing reaction forces (from Section 4.1) were applied to the bearing housing surfaces.
- **Mesh:** A fine tetrahedral mesh was used, with refinement applied at the bearing seats and stress concentration zones (fillets).

2.14.4 4. Von Mises Stress Analysis

The equivalent (Von Mises) stress distribution is shown below. The analysis focuses on ensuring the maximum stress remains below the material's yield strength.

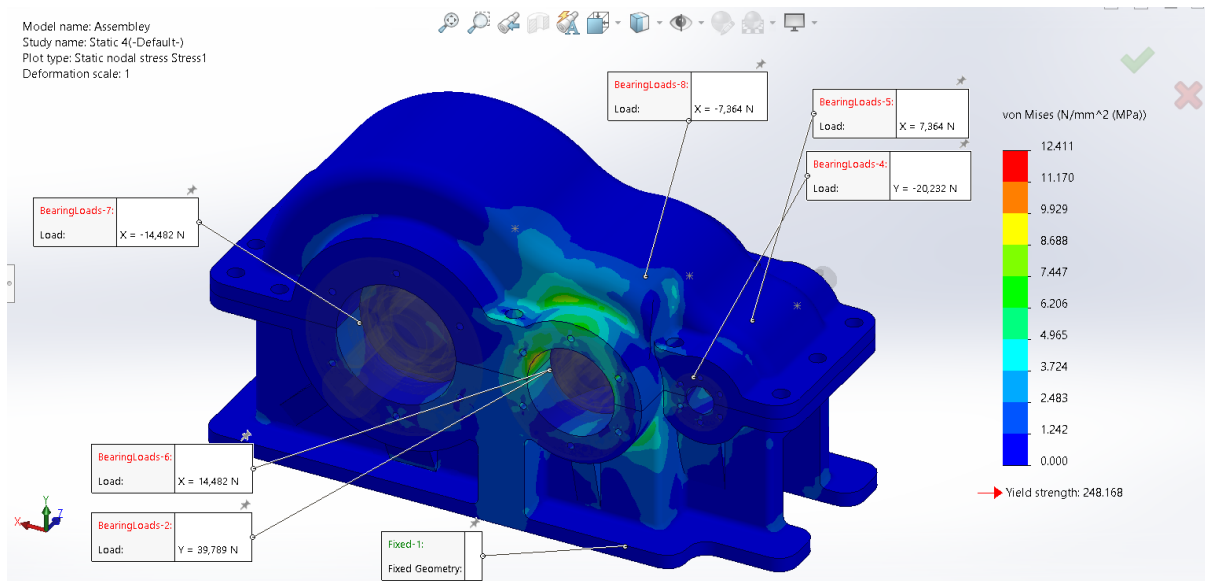


Figure 13: Von Mises Stress Distribution on Gearbox Housing

Safety Factor Calculation: The maximum observed stress (σ_{max}) occurs at the ribbing between the output and intermediate shaft bearing seats.

$$\sigma_{max,FEA} = 15 \text{ MPa} \quad (67)$$

$$n_{housing} = \frac{S_y}{\sigma_{max,FEA}} = \frac{200 \text{ MPa}}{15 \text{ MPa}} = 13 \quad (68)$$

Since $n_{housing} > 3.0$ (standard for cast housings), the design is verified to be safe against static failure.

2.14.5 5. Deformation and Stiffness Check

Excessive deformation in the housing can lead to shaft misalignment, causing edge loading on gears and premature bearing failure.

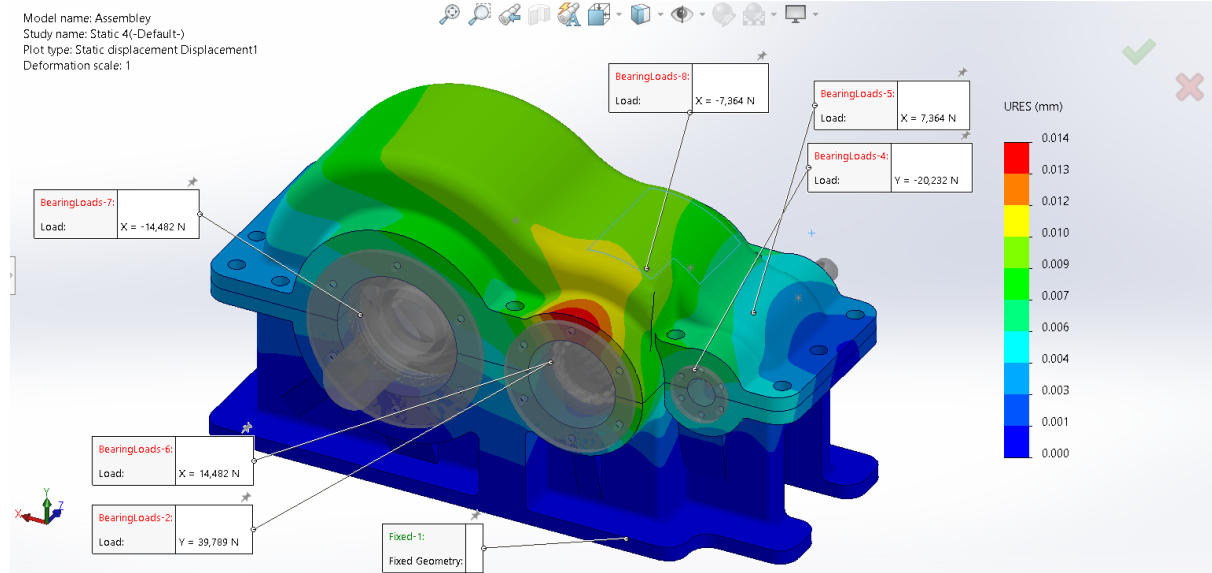


Figure 14: Total Deformation of Gearbox Housing

Misalignment Verification: The maximum total deformation (δ_{max}) was observed at the intermediate shaft bearing seat.

$$\delta_{max} = 0.014 \text{ mm} \quad (69)$$

The allowable deflection for gear mesh alignment is typically limited to 0.05 mm per 100 mm of span. Given the maximum deflection is extremely low ($\delta_{max} < 0.05 \text{ mm}$), the housing provides sufficient stiffness to maintain proper gear and bearing alignment under full load.

3. Conclusion

The objective of this project was to design, analyze, and validate a heavy-duty double reduction gearbox capable of transmitting **175 kW** of power with a reduction ratio of **1:10**. The design process integrated classical machine design principles with modern Computer-Aided Drawing and Engineering (CAD/CAE) tools to ensure reliability, manufacturability, and safety under continuous industrial operation.

This report has detailed the complete engineering workflow, from the initial kinematic arrangement to the final Finite Element Analysis (FEA) of the housing. The following subsections summarize the critical findings and the validation of the system's integrity.

3.1 Transmission System Integrity

The decision to utilize **Spur Gears** manufactured from **ASTM A148 Grade 80-50 Cast Steel** was driven by the specific requirement to utilize local casting capabilities. While spur gears typically generate higher noise levels than helical counterparts, they offer distinct advantages for this high-torque application:

- **Elimination of Axial Thrust:** The lack of axial forces simplified the bearing selection process, allowing for the use of standard deep groove ball bearings rather than complex tapered roller assemblies.
- **Structural Safety:** The analytical stress calculations, based on the Lewis and Hertzian equations, yielded bending safety factors of **2.42** for the high-speed stage and **3.00** for the low-speed stage. These values provide a sufficient margin to account for potential shock loads during startup.

3.2 Shaft and Bearing Reliability

The shaft design was a critical focus area due to the combined loading conditions (torsion from power transmission and bending from gear separation forces).

- **Shaft Strength:** The selection of **AISI 1045 Cold Drawn Steel** proved effective. The distortion energy theory (Von Mises) analysis confirmed that the shafts operate well within the elastic region, with a minimum factor of safety of **2.46** under maximum load.
- **Bearing Longevity:** A conservative approach was taken in bearing selection. By utilizing **SKF Explorer Series** bearings (6312, 6318, 6324), the design achieved a minimum rated life (L_{10h}) of over **10,000 hours** for the high-speed shaft and over **45,000 hours** for the output shaft. This ensures maintenance intervals align with standard industrial schedules.

3.3 Structural Rigidity of the Housing

The gearbox housing acts as the backbone of the system. The use of **Grey Cast Iron (ASTM A48 Class 30)** provided the necessary vibration damping properties essential for a spur gear setup.

- **Stiffness Validation:** The FEA simulation highlighted the effectiveness of the ribbing structure. The maximum deflection observed was only **0.014 mm**, which is significantly below the permissible limit of 0.05 mm. This rigidity guarantees that the gear mesh remains perfectly aligned, preventing edge loading and premature tooth wear.
- **Over-design for Safety:** The high safety factor of **13** for the housing suggests it is extremely robust. While this might appear over-engineered, the added mass contributes to thermal inertia and noise attenuation, which are beneficial for this specific application.

3.4 Recommendations for Operation and Maintenance

While the mechanical design is sound, the operational longevity of the gearbox will depend on adherence to the following recommendations:

1. **Lubrication Management:** Given the high power throughput (175 kW) and the sliding contact of spur gears, the external forced cooling system is mandatory. The oil temperature should be monitored to remain below 80°C to maintain film thickness.
2. **Initial Run-in Period:** A "soft start" period is recommended for the first 50 hours of operation to allow the cast gear teeth to bed in properly, improving surface finish and reducing noise.
3. **Vibration Monitoring:** Due to the impulsive nature of spur gear engagement, periodic vibration analysis is suggested to detect early signs of bearing wear or tooth misalignment.